

Starting to use *Mathematica* is about as easy as starting to use an electronic calculator. You type in your calculations, and *Mathematica* gives you back the results. The main thing you have to learn is how to describe your calculations in *Mathematica*'s language. Most of the time, you will find this language close to what you are used to from mathematics. However, not all the notations and conventions of mathematics can conveniently be used on a computer. As one simple example, many mathematical symbols do not appear on most computer keyboards. You will have to learn how to translate your calculations into *Mathematica*'s language.

This chapter starts by describing how to do numerical calculations with *Mathematica*. It shows you how to use *Mathematica* essentially like an electronic calculator. Once you have read the first few subsections of the chapter, you should be able to start making productive use of *Mathematica*. You will already have seen most of the basic elements that you need to do all kinds of calculations with *Mathematica*.

As you progress through the chapter, you will learn about many of the things that *Mathematica* can do. You will see how to do algebra with *Mathematica*, how to plot graphs, and how to create your own functions. By the time you have finished the chapter, you should have the working knowledge of *Mathematica* that you will need to do many kinds of practical problems.

What you see in this chapter is nevertheless in many respects just the surface of *Mathematica*. Underlying all the various features and capabilities described in this chapter, there is a powerful and general mechanism. Much of the time, you will never need to know about this mechanism. However, if you want to get the most out of *Mathematica*, you will have to understand it. Chapter 2 describes the structure of *Mathematica*, and gives you a more global picture of the system.

This chapter is a self-contained introduction to *Mathematica*. You should be able to read it even if you have never used a computer before. (You should look at the documentation that came with your copy of *Mathematica* to find out about the special features of your particular computer, and about using *Mathematica* on it.)

Most of the material in this chapter assumes no knowledge of mathematics beyond high-school level. The methods *Mathematica* uses to do calculations are often highly sophisticated, but that does not mean that you need to use any advanced mathematics in describing your calculation to *Mathematica*. Some parts of this chapter venture into slightly more complicated mathematics; if you do not understand it, skip that subsection – it will not be used elsewhere in the chapter. If you want to know the full mathematical details, try looking in Chapter 3.

About the illustration overleaf:

This picture was produced by the *Mathematica* command

`Plot3D[Sin[x y], {x, 0, Pi}, {y, 0, Pi}]` described in Section 1.8. The option setting `PlotPoints->30` was used.

■ Running *Mathematica*

The details of installing and running *Mathematica* vary from one computer system to another.

If *Mathematica* came with your computer system, then it should be ready to use as soon as you have set up your computer.

If you bought *Mathematica* separately, then it should have come with installation instructions. The instructions are probably quite simple, but you should make sure to carry them out carefully and completely.

For details of how to run *Mathematica* on your particular computer system, you should look at the on-line documentation. On systems with standard terminal interfaces, the documentation is usually in a file called **README**. On systems with graphical interfaces, you should be able to get to on-line documentation by selecting the **About Mathematica** menu item.

Although the details of running *Mathematica* differ from one computer system to another, the structure of *Mathematica* calculations is the same in all cases. You enter input, *Mathematica* processes it, and then returns a result.

Throughout this book, “dialogs” with *Mathematica* are shown in the following way:

The computer prints <code>In[1]:=</code> . You just	<code>In[1]:= 2 + 2</code>
type in <code>2 + 2</code> . The line that starts with	<code>Out[1]= 4</code>
<code>Out[1]=</code> is the result from <i>Mathematica</i> .	

Such dialogs may look slightly different on your computer. In particular, the `In[1]:=` may be inserted only *after* you type in your input.

When you read through this book, you should try out some of the examples on your own *Mathematica* system. Here are a few points to watch:

- Until you are familiar with *Mathematica*, make sure to type the input *exactly* as it appears in the book. Do not change any of the capital letters or brackets. Later, you will learn what things you can change. When you start out, however, it is important that you do not make any changes, otherwise you may not get the same results as in the book.
- You will see that the lines in each dialog are numbered in sequence. Most subsections in the book contain separate dialogs. To make sure you get exactly what the book says, you should start a new *Mathematica* session each time the book does.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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